

Chapter 4 — String Length & Basic Operations



SLIDE 1 — Finding String Length

What is String Length?

The **length of a string** is the number of characters present in it.

For example:

```
char word[] = "HELLO";
```

The string contains:

H	E	L	L	O
1	2	3	4	5

So,

Length = 5 

 **Think of it like this:**

Imagine counting people in a line 

				
1	2	3	4	5

String length works in the same way — **count how many characters are there.**

 **Remember**

String Length = Number of characters in the string

 **'\0'** marks the end of a string, but it is **not counted as part of the string's length.**



SLIDE 2 — Using `strlen()`



What is `strlen()`?

`strlen()` is a function used to find the length of a string.

It comes from the:

```
#include <string.h>
```



Example

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char word[] = "HELLO";
```

```
    printf("%d", strlen(word));
```

```
    return 0;
```

```
}
```

Output:

5



What is happening?

word → "HELLO"

↓

strlen()

↓

5

`strlen()` counts the characters and gives the length.

Memory Trick

strlen = String Length 

Just remember:

str → string

len → length

SLIDE 3 — Counting Characters

 What does “Counting Characters” mean?

Sometimes we don't just want the total length.

We may want to **count how many characters satisfy a condition**.

For example:

"BANANA"

How many **A**s are there?

B	A	N	A	N	A
	↑		↑		↑

 There are **3 A's**.

Basic idea

We can check each character one by one:

```
char word[] = "BANANA";  
int count = 0;
```

Then:

 Check character



Is it 'A'?



YES → count++



Move to next character

 **Remember**

Check → Match → Count

This idea will help us when counting specific characters. 

SLIDE 4 — Searching for a Character

 **What does “Search” mean?**

Searching for a character means checking whether a **particular character exists inside a string**.

Example:

```
char word[] = "HELLO";
```

Let's search for:

'L'

C checks:

H → 

E → 

L →  FOUND!

Simple idea

```
char word[] = "HELLO";
```

```
for(int i = 0; word[i] != '\0'; i++)  
{  
    if(word[i] == 'L')  
    {  
        printf("Found");  
    }  
}
```

What is happening?

Character → Compare → Match?

↓
YES → Found 

Memory Trick

SEARCH = CHECK each character until you find a match 

SLIDE 5 — Counting Specific Characters + Recap

Counting a Specific Character

Now let's combine **checking + counting**.

Example:

```
char word[] = "BANANA";  
int count = 0;
```

```
for(int i = 0; word[i] != '\0'; i++)
```

```
{
    if(word[i] == 'A')
    {
        count++;
    }
}
```

printf("%d", count);

Output:

3

🧩 **What happened?**

B → ❌

A → ✅ count = 1

N → ❌

A → ✅ count = 2

N → ❌

A → ✅ count = 3

🧠 CHAPTER 4 RECAP

📏 **String Length** → Number of characters

🔧 **strlen()** → Finds string length

📊 **Counting Characters** → Count characters that match a condition

🔍 **Searching** → Check whether a character exists

🎯 **Counting Specific Characters** → Find and count a particular character

💡 **BIG MEMORY TRICK**

LENGTH → COUNT → SEARCH → COUNT AGAIN 🎯

🏆 **LEVEL CLEARED!**

You can now **measure, search, and count characters inside a string.**

QUESTIONS:

Q1. MCQ – String Length

What is the length of the following string? 

```
char word[] = "CODING";
```

- ♦ A) 5
- ♦ B) 6
- ♦ C) 7
- ♦ D) 8

 **Hint:** Count only the characters in "CODING".

Glossary:

-  **String length** – The number of characters in a string.

 **Answer:** B) 6

Q2. Fill in the Blank – `strlen()`

Complete the code to find the length of the string.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char word[] = "HELLO";
```

```
    printf("%d", _____(word));
```

```
    return 0;  
}
```

 **Hint:** Which function is specifically used to find **string length**?

 Glossary:

-  **String length** – Number of characters in a string.
-  **Function** – A block of code that performs a particular task.

 **Answer:** `strlen`

 Q3. Code Choice – Searching

Which code correctly searches for the character 'A' in a string? 

```
char word[] = "BANANA";
```

♦ **A**

```
if(word[i] = 'A')
```

♦ **B**

```
if(word[i] == 'A')
```

♦ **C**

```
if(word == 'A')
```

◆ D

```
if(word[i] != 'A')
```

💡 **Hint:** To check whether two values are equal, use the **comparison operator**.

📖 **Glossary:**

- 🔍 **Search** – Check whether something exists.
- ⚖️ **Compare** – Check whether two values are equal or different.

✅ **Answer: B**

Q4. Fill in the Blank – Counting Characters

Complete the code to count how many 'A' characters are present in "BANANA".

```
char word[] = "BANANA";  
  
int count = 0;  
  
for(int i = 0; word[i] != '\0'; i++)  
{  
    if(word[i] == 'A')  
    {  
        -----;  
    }  
}
```

```
}
```

```
printf("%d", count);
```

💡 **Hint:** Whenever 'A' is found, increase `count` by 1.

📖 **Glossary:**

-  **Count** – The number of times something occurs.
- **+** **Increment** – Increase a value.

✅ **Answer:** `count++`

🎯 Q5. Code Choice – Counting Specific Characters

What will be the output? 🤔

```
char word[] = "APPLE";  
  
int count = 0;  
  
for(int i = 0; word[i] != '\0'; i++)  
{  
    if(word[i] == 'P')  
    {  
        count++;  
    }  
}  
  
printf("%d", count);
```

- ♦ A) 1
- ♦ B) 2
- ♦ C) 3
- ♦ D) 5

💡 **Hint:** Count how many times P appears in "APPLE". 🔍

📖 **Glossary:**

- 🎯 **Specific character** – A particular character being searched or counted.
- 📊 **Count** – Number of times something occurs.

✅ **Answer:** B) 2